Notes: De Fiore, Hoerova, Rogers, and Uhlig (RES, 2026)

Money Markets, Collateral, and Monetary Policy

Contents

1 Summary

De Fiore, Hoerova, Rogers, and Uhlig (RES, 2026) study how disruptions in euro-area money markets during the Global Financial Crisis and European Sovereign Debt Crisis affected the real economy, and how the ECB's collateralized lending policy mitigated those effects. The paper links four empirical facts - the collapse of unsecured interbank lending, the rise of secured lending, the sharp increase in private market haircuts on southern sovereign collateral, and the large rise in ECB borrowing by southern banks - to a quantitative general-equilibrium banking model.

The central economic mechanism is **collateral scarcity**. Banks need short-term funding to manage deposit outflows. Some banks can borrow unsecured, while others must borrow in secured markets and post sovereign bonds as collateral. When more banks lose access to unsecured funding, or when haircuts on sovereign collateral rise, banks must hold more collateral and reserves. This crowds out productive capital investment. Since banks are leveraged capital investors, the fall in investment lowers bank net worth and amplifies the downturn.

The model separates two regions: the **North** and the **South**. Both have banks and sovereign bonds, but southern bonds experience a sharp haircut shock during the sovereign debt crisis. This shock directly tightens collateral constraints for southern banks. It also spills over to northern banks because southern banks rebalance toward reserves to manage liquidity, reducing the reserve supply available to northern banks. Northern banks then rely more on bond collateral, raising collateral premia and crowding out capital investment across the monetary union.

The policy comparison is between two central-bank regimes. Under the benchmark **CO policy**, the central bank provides collateralized loans to banks at stable and favourable haircuts. Under the alternative **CB policy**, the central bank keeps its balance sheet constant and does not provide additional collateralized lending. The paper finds that the fall in output, investment, and capital would have been roughly twice as large under the constant-balance-sheet policy.

2 Big picture

- **Question.** How do money-market frictions and collateral haircuts transmit to the macroeconomy, and how much did ECB collateralized lending matter?
- **Empirical setting.** Euro-area interbank money markets during 2003–2015, with a focus on the Global Financial Crisis and the European Sovereign Debt Crisis.
- **Key facts.** Unsecured interbank borrowing collapsed, secured borrowing rose, private haircuts on southern sovereign bonds increased sharply, southern banks borrowed heavily from the ECB, and household deposits moved inversely with ECB funding.
- **Model innovation.** A dynamic GE model with heterogeneous banks, unsecured and secured interbank markets, sovereign-bond collateral, endogenous collateral scarcity, and a central bank.
- **Main mechanism.** More secured funding needs and higher haircuts increase collateral demand; banks substitute collateral and reserves for capital; capital falls; bank net worth falls; leverage amplifies the real effects.
- **Policy result.** Favourable ECB haircuts on collateralized lending create a haircut gap relative to private markets, easing collateral scarcity and preventing a larger decline in output, investment, and capital.

3 Empirical facts

The paper begins by documenting four money-market developments.

1. **Decline of unsecured borrowing.** The unsecured segment of the interbank market dried out across both North and South. Banks substituted toward secured borrowing.
2. **Rise in southern collateral haircuts.** During the sovereign debt crisis, private haircuts on southern government bonds increased substantially. This reduced the borrowing capacity of banks using those bonds as collateral.
3. **Rise in ECB borrowing.** Southern banks increased borrowing from the ECB roughly eightfold, while the ECB kept accepted-collateral haircuts nearly unchanged.
4. **U-shaped household deposits.** Household funding fell when central-bank borrowing rose and recovered when central-bank borrowing waned, especially in the South.

These facts are important because they jointly identify the paper's core margin: banks moved away from trust-based unsecured funding and toward collateral-intensive funding, while the central bank became a substitute collateralized lender when private collateral terms deteriorated.

4 Model structure

The economy has households, firms, governments, a central bank, and heterogeneous banks. The model period has a **morning** and an **afternoon**. In the morning, banks choose portfolios of capital, sovereign bonds, reserves, deposits, and central-bank borrowing. In the afternoon, depositors reshuffle deposits, creating liquidity outflows for some banks and inflows for others.

Banks differ along two dimensions. First, a bank can be **connected** or **unconnected**. Connected banks can borrow unsecured in the afternoon. Unconnected banks must borrow against collateral in the secured market. Second, banks are located in the North or the South and hold home-biased sovereign collateral.

The key constraints are:

- a **leverage constraint**, which limits bank assets relative to bank value and net worth;
- a **central-bank collateral constraint**, which ties ECB funding to the collateral value of pledged sovereign bonds;
- an **afternoon liquidity constraint**, which requires unconnected banks to carry enough reserves or unpledged collateral to meet deposit outflows.

The paper shows that the leverage constraint and the afternoon constraint are both essential. Without either constraint, changes in the connected-bank share or private collateral haircuts would not have aggregate effects.

5 Mechanism

The model's mechanism can be read through four forces.

- **Capital crowding out.** When banks need to hold more collateral assets, they invest less in productive capital.
- **Bond-reserves substitution.** When bond collateral becomes costly, banks may switch from bonds to reserves to manage liquidity.
- **North-South liquidity spillover.** Southern haircut shocks make southern banks hoard reserves, which creates collateral scarcity for northern banks as well.
- **Haircut gap.** When private haircuts rise but central-bank haircuts remain stable, central-bank loans become attractive and ease collateral scarcity.

The haircut gap is the central policy lever. It allows banks to substitute central-bank collateralized funding for private secured funding precisely when private collateral terms become restrictive. This creates additional reserves and weakens the capital-crowding-out force.

6 Policy experiment

The benchmark policy is **collateralized operations** (CO). In this regime, the central bank lends reserves to banks against collateral at a favourable haircut, with $\eta = 0.97$. The counterfactual is a **constant balance sheet** (CB) regime, with no additional collateralized lending and $\eta = 0$.

The counterfactual shows that central-bank lending mattered quantitatively. With no additional balance-sheet expansion, the adverse impact of the southern haircut shock on output, investment, and capital would have been much larger. The paper's main quantitative result is that the fall in these real variables would have been approximately twice as high under the alternative policy.

7 How to use this paper

This paper is useful for thinking about monetary policy transmission in environments where central-bank balance-sheet policies affect the value of collateral. It also provides a framework for linking micro-level funding-market disruptions to macro-level outcomes.

For related research, the paper is especially useful for:

- bank liquidity management and interbank networks;
- repo markets, collateral haircuts, and sovereign collateral;
- ECB crisis policies and central-bank balance sheets;
- financial frictions and real investment;
- general-equilibrium effects of secured versus unsecured funding;
- cross-region spillovers through collateral scarcity.

8 Notes-created summary tables

8.1 Empirical observations and model interpretation

Observation	Model interpretation
Unsecured borrowing declines	A lower share of connected banks forces more banks into collateralized funding.
Secured borrowing rises	Banks substitute away from trust-based unsecured markets toward collateral-backed markets.
Southern private haircuts rise	Southern sovereign collateral loses borrowing capacity, tightening southern bank liquidity constraints.
Southern ECB borrowing rises	Favourable central-bank haircuts make ECB funding attractive relative to private secured funding.
Household deposits fall and recover	Banks substitute between deposit funding and central-bank funding as money-market conditions change.

8.2 Main mechanisms

Mechanism	Economic logic
Capital crowding out	Collateral and reserves absorb bank balance-sheet capacity that otherwise could finance productive capital.
Bond-reserves substitution	Reserves can be a substitute liquidity asset when bond collateral becomes costly.
North-South spillover	Southern reserve demand can reduce reserve availability for northern banks, forcing them to hold more collateral.
Haircut gap	The difference between private and central-bank haircuts makes central-bank funding a crisis backstop.
Leverage amplification	Lower capital investment and higher collateral premia reduce bank value and net worth, tightening bank balance sheets further.

9 Direct paper exhibits

9.1 Title, abstract, and core facts

Money Markets, Collateral, and Monetary Policy

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We document dramatic changes in euro area interbank money markets during the financial and sovereign debt crises: unsecured borrowing declined across the euro area, while secured market haircuts on sovereign bonds increased, and bank borrowing from the European Central Bank rose in southern countries. We construct a quantitative general equilibrium model to assess the macroeconomic impact of these developments and the associated policy response. Our model features heterogeneous banks and sovereign bonds, secured and unsecured money markets, and a central bank. We compare a benchmark policy—the central bank providing collateralized lending to banks at haircuts lower than the market—with an alternative policy that maintains a constant central bank balance sheet. We show that the fall in output, investment, and capital would have been twice as high under the alternative policy.

Key words: Money markets, Collateral, Monetary policy, Central bank balance sheet

JEL codes: E44, E52, E58

Title page and abstract

Euro area money markets underwent dramatic changes during the Global Financial Crisis and the European Sovereign Debt Crisis. We start by documenting four interrelated facts in particular. First, activity in the segment of the unsecured money market, where banks borrow and lend to each other based on trust, progressively dried out. The share of unsecured interbank transactions declined throughout the euro area, with banks substituting towards secured transactions, where borrowers must secure the loan by pledging enough securities as collateral. Second, with the onset of the European Sovereign Debt Crisis in 2010, collateral haircuts—the difference between the security price and its value as collateral—on government bonds from countries in the southern euro area increased substantially, reducing the amount banks could borrow against pledged assets.[1] Third, during this period, banks in the southern countries increased secured borrowing from the European Central Bank (ECB) 8-fold. This occurred while the ECB kept haircuts on accepted collateral nearly unchanged and well below those set in the market, creating a favourable "haircut gap" between private and ECB haircuts. Fourth, household deposits exhibited a U-shaped pattern: they fell when banks borrowed from the central bank, and vice versa when the central bank borrowing waned. This pattern was more pronounced in the South, where the ECB borrowing take-up was larger.

Our goal in this article is to understand how these money market developments impacted the real economy and the role played by policies employed by the ECB. We develop a dynamic general equilibrium model with the banking sector at its heart to lay out the mechanism and quantify the effects. As in Gertler and Karadi (2011) and Gertler and Kiyotaki (2010), banks are capital investors who can leverage their net worth by issuing deposits up to some limit. Similar to Bianchi and Bigio (2022), banks must also manage short-term deposit outflows using the interbank market. Our key innovation here is to embed additional empirically relevant features of money markets: some banks can borrow unsecured while others only have access to secured funding, which requires posting collateral, typically government bonds. How much secured funding the bank can access depends on the asset's collateral value: the higher the asset value and the lower the haircut, the larger the available funding. In line with the euro area context, our model features one central bank but heterogeneous government bonds. Specifically, there are two regions, the "North" and the "South" each with its own government bonds and a banking sector that holds a home-biased portfolio of the bonds.

Our model, which we calibrate to euro area data, indicates that the observed money market

Four facts, model mechanism, and macro impact

9.2 Empirical facts: interbank markets, haircuts, and deposits

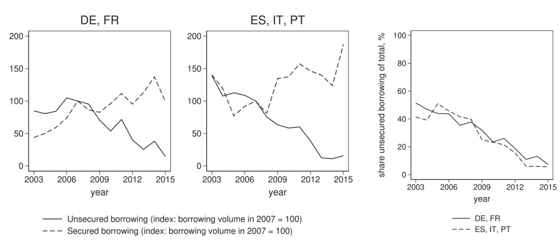

FIGURE 1
Unsecured and secured interbank borrowing

Notes: The two figures on the left present cumulative quarterly turnover of unsecured and secured interbank borrowing over 2003–15 (the time span of the data collection), for northern (DE and FR) and southern (ES, IT, and PT) regions, respectively. Turnover is defined as the sum of all borrowing transactions over the second quarter of each year (the quarter in which the data were collected), reported by banks participating in the ECB's money market survey (MMS) and normalized to 100 in 2007. The MMS panel comprised 98 euro area banks. The figure on the right presents the relative share of unsecured borrowing in total, for DE and FR (solid line) and for ES, IT, and PT (dashed line). Source: Euro Area Money Market Survey.

Figure 1: unsecured and secured interbank borrowing

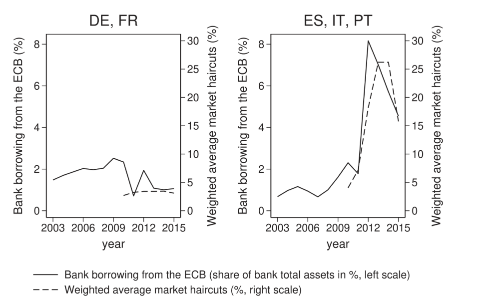

FIGURE 2
Bank borrowing from the ECB as % of total assets, and market haircuts

Notes: The figure presents bank borrowing from the ECB as % of bank total assets (solid line, left scale in each figure) and weighted average market haircuts in % (dotted line, right scale in each figure) over 2010–5, for northern (DE and FR) and southern (ES, IT, and PT) regions, respectively. Region-level haircuts are obtained as simple averages across maturities under 30 years. Region-group-level haircuts are constructed as weighted averages across the northern (southern) regions, with weights given by the shares of the respective banking sector assets in total in 2010 (the weights are 48% and 52% for DE and FR, and 49%, 44%, and 7% for IT, ES, and PT, respectively). Source: ECB and LCH.Clearnet

Figure 2: ECB borrowing and market haircuts

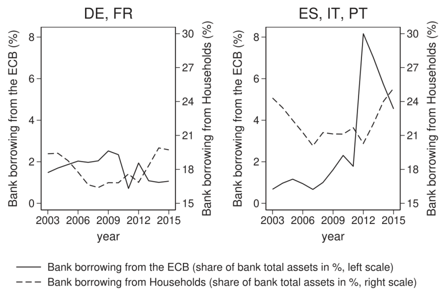

FIGURE 3
Bank borrowing from households and the ECB, % of total assets

Notes: The figure presents bank borrowing from the ECB (solid line, left scale in each figure) and from households (dotted line, right scale in each figure) as % of bank total assets over 2003-2015, for northern (DE and FR) and southern (ES, IT, and PT) regions, respectively. Source: ECB

Figure 3: household funding and ECB borrowing

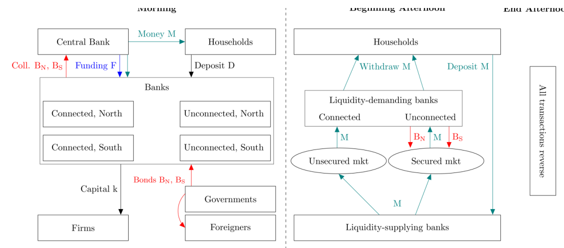

FIGURE 4
Model overview

Figure 4: model overview and timing

9.3 Model forces and constraints

private sector and central bank haircuts is sufficiently large.

This interplay of trade-offs and market interactions generates the main forces that help our model explain the data. Aside from the impact of leverage well known from models building on Gertler and Kiyotaki (2010) and Gertler and Karadi (2011), the market for collateral (bonds, money) gives rise to four forces that are novel to the analysis here.

(1) **Capital Crowding Out:** suppose the banking sector needs to hold more collateral assets. This can be due to a higher share of unconnected banks or to a lower quality of existing collateral (*e.g.* higher private haircuts on bonds). As banks have limits on leverage, this crowds out productive capital investment.

(2) **Bond-Reserves Substitution:** when there is an increase in collateral scarcity, banks pay a higher *collateral premium* in equilibrium, *i.e.* they forego more interest investing in bonds relative to capital. If the premium is high enough, banks may find it optimal to switch from bonds to reserves to manage liquidity. This helps mitigate the loss in bank profitability.

(3) **North–South Liquidity Spillover:** when both regions hold reserves, there are strong spillovers of collateral scarcity across regions. For example, suppose that the South experiences a drop in the collateral quality of its bonds. In response, unconnected banks in the South rebalance to reserves as a way to manage liquidity. However, the reserves must come from other banks unless the central bank steps in, *i.e.* unconnected banks in the North, which, therefore, must rebalance from reserves to collateral assets for liquidity needs. As a result, banks in the North end up facing a collateral scarcity issue as well.

(4) **Haircut Gap:** the gap between the private market and the central bank haircut is what we call the "haircut gap". If the gap is large enough, banks will start to use the central bank as a source of funding since high private haircuts make managing afternoon deposit flows costly. In addition, by lending new reserves, the central bank increases the aggregate supply of reserves for banks. This reduces collateral scarcity across the monetary union by alleviating collateral shortages.

Four model forces

Finally, the flow budget constraint of the central bank is given by

$$\overline{M}_t - \overline{M}_{t-1} = S_t + Q_t^F \overline{F}_t + Q_t^N B_{N,t}^C + Q_t^S B_{S,t}^C \\ -R_t^N Q_{t-1}^N B_{N,t-1}^C - R_t^S Q_{t-1}^S B_{S,t-1}^C - \overline{F}_{t-1}, \tag{14}$$

where the terms related to the central bank loans to banks, $\bar{F}_t$ and $\bar{F}_{t-1}$, are as explained in the previous paragraph and where R_t^j is the nominal return from holding government bonds of region $j \in \{N, S\}$ defined in equation (9). S_t is the seigniorage of the central bank, which is the net proceeds from asset allocations that are transferred to the government.[18]

We focus on the impact of a monetary policy operation allowing banks to borrow from the central bank at favourable haircuts during crises. Therefore, we compare two policies: benchmark (CO) versus alternative (CB). The idea here is that the benchmark describes collateralized lending operations during the great financial crisis (GFC) and the european sovereign debt crisis, whereas the alternative reflects monetary policy pre-GFC. In particular:

CO Policy (*Collateralized operations*) : $\eta = 0.97$

CB Policy (*Constant Balance Sheet*) : $\eta = 0$

Under CO policy, the central bank haircut is set constant at the pre-GFC private market level of 3% or $1 - \eta = 0.03$. This can become quite an attractive policy for banks if private haircuts increase substantially. In this case, banks may find it optimal to obtain collateralized loans from the central bank so that $\bar{F}_t$ can exceed zero. In contrast, under the CB Policy, banks have no incentive to borrow from the central bank as their collateral has zero value. In this case, the size of the central bank balance sheet is constant at $\bar{F}_t = 0$. Regarding the interest rate on central bank loans, in our numerical analysis, we set it equal to the average interest rate on central bank loans that prevailed over the 2008–14 period (*i.e.* during and after the GFC) in the euro area (see Section 5 for details).

Central-bank policy rules and CO/CB comparison

Knowing their network type and region, banks then choose their portfolio of assets and liabilities and pay dividends. We assume that dividends are a constant fraction ϕ of net worth. The bank collects nominal deposits ($D_{i,t}$) and accesses nominal secured loans from the central bank at face value ($F_{i,t}$). The ex-dividend net worth plus liabilities are then allocated to capital ($k_{i,t}$), nominal bonds ($B_{i,t}$), and nominal reserves ($M_{i,t}$).

The bank's balance-sheet constraint at the end of the morning is, therefore, given by

$$P_t Q_t^k k_{i,t} + Q_t^j B_{i,t} + M_{i,t} + \phi N_{i,t} = D_{i,t} + Q_t^F F_{i,t} + N_{i,t}, \tag{15}$$

Banks face a variety of additional constraints on their portfolio choice in the morning. Firstly, the collateral constraint at the central bank requires loans not to exceed the value of the bonds pledged, adjusted by the central bank's haircut. In equilibrium, banks will pledge just enough collateral to make the collateral constraint at the central bank bind. This is because any extra bonds can instead be used as collateral in the secured money market in the afternoon, or alternatively, the funds used to buy extra bonds can instead be invested in at least as profitable capital in the morning. We impose that the "just enough" collateral constraint holds, even if banks are indifferent, *i.e.* for all types of banks,

$$F_{i,t} = \eta_t Q_t^j B_{i,t}^F. \tag{16}$$

Given the face value of central bank loans $F_{i,t}$, the bank obtains central bank funding in the amount of $Q_t^F F_{i,t}$.[19]

Second, bonds pledged at the central bank are constrained to be non-negative but also to not exceed the amount of bonds acquired in the morning,

$$0 \leq B_{i,t}^F \leq B_{i,t}. \tag{17}$$

Third, banks face a leverage constraint. As in Gertler and Kiyotaki (2010) and Gertler and Karadi (2011), we assume there is a moral hazard constraint in that bank managers may run away with a fraction λ of their assets at the end of the morning. Let $V_{i,t}$ be the value of a bank at the end of the morning after the type of the bank is known, deposit claims are issued, central bank funding is accessed, and assets are purchased, but before dividends are paid. At this point, the banker has the option to walk away with a fraction λ of the assets and transfer them back to the household. They will decide to do so if the value of the bank itself, $V_{i,t}$, is below a fraction λ of the value of the assets held. In this case, creditors will lose out as they can only recover a fraction $(1-\lambda)$ of their investments. As a result, the following leverage constraint must be satisfied for depositors to be willing to deposit with the bank:

Bank balance sheet, collateral, and leverage constraints

Define $\tilde{V}_{\nu,j,t}$ as the value of a bank with average net worth, network type ν, and located in j at the end of the morning, after distributing dividends. We have

$$V_{\nu,j,t} = \phi N_{\nu,j,t} + \tilde{V}_{\nu,j,t}, \tag{28}$$

where

$$\tilde{V}_{\nu,j,t} = \beta E_t \left[\frac{u_c\left(c_{t+1}, h_{t+1}\right)}{u_c\left(c_t, h_t\right)} \frac{P_t}{P_{t+1}} \psi_{t+1} N_{\nu,j,t+1} \right] \tag{29}$$

with net worth $N_{\nu,j,t+1}$, resulting from the investments of period t, given by

$$N_{\nu,j,t+1} = R_{t+1}^k P_t Q_t^k k_{\nu,j,t} + M_{\nu,j,t} + R_{t+1}^j Q_t^j B_{\nu,j,t} - R_t^d D_{\nu,j,t} - R_t^F Q_t^F F_{\nu,j,t} \tag{30}$$

Equation (29) makes use of equation (25) because a bank of type (ν, j) at time t is valued at time $t+1$ at $\psi_{t+1} N_{\nu,j,t+1}$. As a result, (28), (29), and (30) deliver a recursive formulation of (24).

In the morning, after the type is known, a bank of type (ν, j) chooses $k_{\nu,j,t}$, $B_{\nu,j,t}$, $B_{\nu,j,t}^F$, $F_{\nu,j,t}$, $D_{\nu,j,t}$, $M_{\nu,j,t}$ to maximize $V_{\nu,j,t}$, subject to the leverage constraint

$$V_{\nu,j,t} \geq \lambda \left(P_t Q_t^k k_{\nu,j,t} + Q_t^j B_{\nu,j,t} + M_{\nu,j,t} \right),$$

the budget constraint

$$P_t Q_t^k k_{\nu,j,t} + Q_t^j B_{\nu,j,t} + M_{\nu,j,t} + \phi N_{\nu,j,t} = D_{\nu,j,t} + Q_t^F F_{\nu,j,t} + N_{\nu,j,t},$$

the central bank funding constraints

$$\begin{aligned} F_{\nu,j,t} &\leq \eta_t Q_t^j B_{\nu,j,t}^F \\ 0 &\leq B_{\nu,j,t} - B_{\nu,j,t}^F, \end{aligned}$$

the non-negativity constraints

$$M_{\nu,j,t} \geq 0, B_{\nu,j,t} \geq 0, F_{\nu,j,t} \geq 0, D_{\nu,j,t} \geq 0, k_{\nu,j,t} \geq 0, \tag{31}$$

and the afternoon constraint for unconnected banks

$$\omega^{\max} D_{\nu,j,t} \leq M_{\nu,j,t} + \tilde{\eta}_t^j Q_t^j \left(B_{\nu,j,t} - B_{\nu,j,t}^F \right). \tag{32}$$

The problems above are linear programming problems, maximizing a linear objective subject to linear constraints. So, the solution is either a corner solution or there will be indifference between certain asset classes, resulting in no-arbitrage conditions. The optimality conditions are reported in Supplementary Appendix E. The definition of the equilibrium and the full system of the equilibrium conditions are reported in Supplementary Appendices F and G, respectively.

Aggregated bank problem and constraints

4.2.1. A model with no leverage constraint. Suppose there is no leverage constraint for any bank, but unconnected banks face an afternoon constraint. The following proposition ensures that shocks to $(\xi, \tilde{\eta}^j)$ do not matter.

Proposition 1 (no leverage constraint). *Consider the model described in Section 3 but without a leverage constraint. Consider two values for the sequence of parameters* $(\xi_t, \tilde{\eta}_t^N, \tilde{\eta}_t^S)$, *indexed by A and B, and fix all other parameters. If there is an equilibrium for* $(\xi_{t;A}, \tilde{\eta}_{t;A}^N, \tilde{\eta}_{t;A}^S)$, *then there is an equilibrium for* $(\xi_{t;B}, \tilde{\eta}_{t;B}^N, \tilde{\eta}_{t;B}^S)$, *where all aggregate quantities are the same.*

The proof is in Supplementary Appendix I. Intuitively, without a leverage constraint, connected banks are fully unconstrained, so they invest in capital until all risk-adjusted returns of assets and liabilities held are the same. Consequently, this model is isomorphic to households investing in capital themselves directly without any banks: shocks to characteristics of the interbank market are irrelevant.

4.2.2. A model without afternoon constraint. Suppose that banks in region j face a leverage constraint but that there is no afternoon constraint for unconnected banks. The following proposition ensures that shocks to $(\xi, \tilde{\eta}^j)$ do not matter.

Proposition 2 (no afternoon constraint). *Consider the model described in Section 3 but without an afternoon constraint for unconnected banks. Then the equilibrium is independent of any shocks to* $(\xi, \tilde{\eta}^j)$.

Intuitively, without the afternoon constraint, the two bank types are identical. Therefore, any movement in ξ does not affect the equilibrium because adjusting the shares between identical banks has no impact. In addition, $\tilde{\eta}^j$ disappears from the optimization problem of banks. Banks

Propositions: leverage and afternoon constraints are essential

decision on $D_{u,j}$ and $F_{u,j}$ in the morning. As dictated by the afternoon constraint, banks can do so by either using money $(M_{u,j})$ or IOUs collateralized with bonds $(B_{u,j})$. Which of the two the bank holds depends on the "collateral premium" for bonds, Λ^j, equalling[26]

$$\Lambda^j = \frac{R^k - R^j}{\tilde{\eta}^j}. \tag{33}$$

Intuitively, to issue $(1/\omega^{\max})$ of deposits, a bank must hold $(1/\tilde{\eta}^j)$ units of bonds. The cost Λ^j of doing so is the returns forgone from having to invest $(1/\tilde{\eta}^j)$ units in bonds instead of capital. By the same logic, if a bank uses money to back $(1/\omega^{\max})$ deposit units, the returns forgone are $R^k - 1$. There are three cases:

$$M_{u,j} = \begin{cases} 0 & \text{if} \quad \Lambda^j < R_k - 1 \\ \in \left[0, \omega^{\max} D_{u,j}\right] & \text{if} \quad \Lambda^j = R_k - 1 \,; \\ \omega^{\max} D_{u,j} & \text{if} \quad \Lambda^j > R_k - 1 \end{cases} \quad B_{u,j} = \frac{F_{u,j}}{Q^j \eta} + \frac{1}{Q^j \tilde{\eta}^j}\left(\omega^{\max} D_{u,j} - M_{u,j}\right). \tag{34}$$

If $\Lambda^j > R^k - 1$, money is a cheaper source of collateral than bonds. Thus, money is used exclusively to satisfy the afternoon constraint. Conversely, if $\Lambda^j < R^k - 1$, only bonds are used, while if $\Lambda^j = R^k - 1$, then any bond-money mix is optimal. Note also that additional bonds must be held as collateral for any central bank funding, at a haircut η, as bonds are the only acceptable collateral for the central bank.

We can now understand how the bank chooses between the two funding sources: deposits $(D_{u,j})$ and central bank funding $(F_{u,j})$. An additional unit of deposits earns X_d:

$$X_d = R_k - R_d - \omega^{\max} \min\left\{R_k - 1, \Lambda^j\right\}. \tag{35}$$

The first term, $R_k - R_d$, is the return earned if the bank invests the deposit unit in capital. The final term reflects the fact that banks cannot invest all in capital, but must hold collateral to back the deposit unit. This is costly because of the forgone returns, as explained above. The min operator is a result of choosing the least-cost type of collateral: bonds or money.

Conversely, an additional unit of central bank funding earns X_f:

$$X_f = R_k - R_f - \frac{\tilde{\eta}^j}{\eta}\Lambda^j. \tag{36}$$

Here, the collateral cost is without a min operator because bonds must be posted to the central bank. The term $\tilde{\eta}^j/\eta$ is because the central bank applies its own haircut, $1-\eta$, not the private haircut $(1-\tilde{\eta}^j)$, to the bond collateral.

Now, suppose that $\max\{X_d, X_f\} > 0$. The bank will then access funding until the leverage constraint binds because more funding enhances portfolio returns. In other words, $(D_{u,j}, F_{u,j})$ is such that

$$D_{u,j} + F_{u,j} = \frac{V_{u,j}}{\lambda} - (1-\phi)N. \tag{37}$$

Funding choice: deposits versus central-bank funding

9.4 Calibration and shock processes

TABLE 1
Parameter values

Parameter	Description	Value
θ	Capital share in income	0.330
δ	Capital depreciation rate	0.020
β	Discount rate households	0.994
ϵ	Inverse Frisch elasticity	0.400
χ^{-1}	Coefficient in households' utility	0.006
g	Government spending	0.566
κ^{-1}	Average maturity bonds (years)	5.896
ϕ	Fraction net worth paid as dividends	0.025
ξ	Fraction banks with access to unsecured market	0.420
$\tilde{\eta}$	Haircut on bonds set by banks	0.970
η	Haircut on bonds set by central bank	0.970
λ	Share of assets bankers can run away with	0.701
$\omega^{\max}$	Max possible liquidity demand as share of deposits	0.100
α	Intercept foreign demand function	4.764
b^C	Bonds held by central bank	0.968
b^*	Stock of debt	7.443
ϱ	Parameter foreign bond demand	4.097
Q^F	Price central bank loans	0.997

Table 1: parameter values

TABLE 2
Calibration

	Data	Model
Targeted variables		
Govt expenditure/GDP	0.200	0.200
Bank leverage	6.000	6.000
Govt bond spread	0.013	0.013
Share bonds held by banks	0.230	0.230
Share bonds foreign sector	0.640	0.640
Inflation (annual)	0.020	0.020
Non-targeted variables		
CB bond holdings/GDP	0.059	0.085
Govt debt/GDP	0.688	0.657

Table 2: calibration

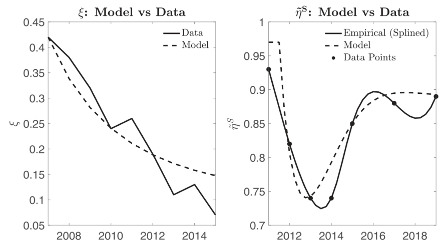

FIGURE 5

Model versus data: evolution of ξ and $\tilde{\eta}^S$

Notes: The figure shows the evolution of, on the left, the share of unsecured borrowing in total, corresponding to the share of connected banks in the model (ξ) and, on the right, the collateral value post-haircut in the South ($\tilde{\eta}^S$). The solid line is the data taken from Section 2.1, while the dashed line is the fitted model-implied evolution of these variables described in Section 5.2. The data points for $\tilde{\eta}^S$ come from the time series evolution of private market haircuts (LCH.Clearnet) on southern government bonds (weighted average across IT, ES, and PT), while the Model line is a spline interpolation through these points that we attempt to fit with an AR(2) process.

Figure 5: model-implied shock processes versus data

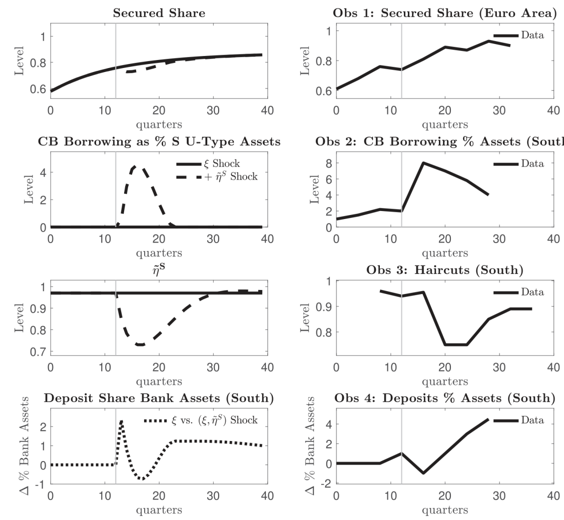

FIGURE 6

Four facts: model versus data

Notes: The figure compares the model outcomes on the first column to their empirical counterparts on the second column regarding the four facts documented in Section 2.1. The first column shows the impulse responses of key variables to exogenous shocks under CO Policy. The timing and process of the shocks in the full experiment are explained in Section 5.2. The solid lines denote the response to the permanent shock to ξ only, which occurs at time 0. The dashed lines in the top three rows denote the response of variables when there is an additional shock to $\tilde{\eta}^S$, which occurs at time $t = 13$. The dashed line in the bottom left figure denotes the response of variables when there is the combined $(\xi, \tilde{\eta})$ shock relative to the ξ shock alone. The timing of the shock to $\tilde{\eta}^S$ is conveyed by the vertical thin grey line. First row: the secured share in the interbank market. Second row: borrowing from central bank as a % of total assets held by unconnected banks in the South. Third row: process for $\tilde{\eta}^S$. Fourth row: deposits as a share of total bank assets in the South. The observations are annual, and the vertical line represents 2011

Figure 6: four facts, model versus data

9.5 Dynamic results and policy counterfactual

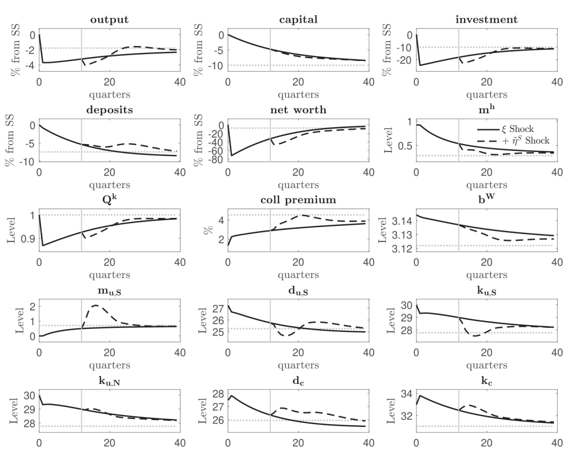

FIGURE 7

Impulse responses: $\xi + \tilde{\eta}$ shock (dashed) versus ξ shock only (solid)

Notes: The figure conveys the impulse responses of key variables to exogenous shocks under CO Policy. The timing and process of the shocks in the full experiment are explained in Section 5.2. The solid lines denote the response to the permanent shock to ξ only that occurs at time 0. The dashed lines denote the response of variables when there is an additional shock to $\tilde{\eta}^S$ that occurs at time $t = 13$. The timing of the shock to $\tilde{\eta}^S$ is conveyed by the vertical thin grey line. The thin dotted horizontal lines indicate the permanent % change/new level of the variable in response to the permanent shock to ξ. First row: % change in aggregate output, capital, and investment. Second row: % change in aggregate deposits, bank net worth, and money held by households. Third row: price of capital, bond collateral premium (as defined in Section 4.3), and world bond demand. Fourth row: money, deposits and capital held by U-types in the South. Fifth row: capital held by U-types in the North, and deposits issued and capital held by connected banks

Figure 7: impulse responses under CO policy

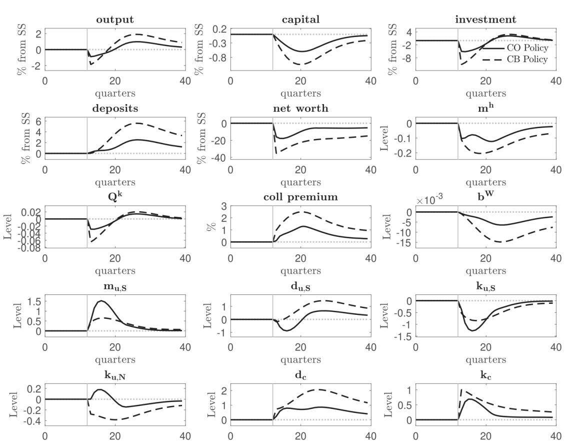

FIGURE 8

Impulse responses: CO policy (solid) versus CB policy (dashed)

Notes: The figure conveys the response of variables in the full experiment compared to a counterfactual where only the ξ shock occurs. In other words, it illustrates the additional effect induced by the shock to $\tilde{\eta}^S$. The timing and process of the shocks in the full experiment are explained in Section 5.2. The solid lines denote the response under CO policy, while the dashed lines denote the response under CB policy. The timing of the shock to $\tilde{\eta}^S$ is conveyed by the thin vertical grey line at $t = 13$. The thin dotted horizontal line is at zero. First row: % change in aggregate output, capital, and investment. Second row: % change in aggregate deposits, bank net worth, and money held by households. Third row: price of capital, bond collateral premium (as defined in Section 4.3), and world bond demand. Fourth row: money, deposits, and capital held by U-types in the South. Fifth row: capital held by U-types in the North, and deposits issued and capital held by connected banks

Figure 8: CO policy versus constant-balance-sheet policy